

Mathematical Calculus

Exam Cheat Sheet

Curious Horizons

Calculus

1. Introduction

Calculus is the mathematical study of continuous change.

Formula

The fundamental theorem of calculus:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Where inline math $F'(x) = f(x)$ represents the derivative.

Exam Trap

Do not forget the constant of integration $+C$ when evaluating indefinite integrals!

Comparison

Concept	Definition	Example
Derivative	Rate of change	$f'(x) = 2x$
Integral	Area under curve	$\int 2x dx = x^2 + C$

- Nested Bullet 1
- Nested Bullet 2

- Nested Bullet 1
- Nested Bullet 2